

Claudespeak: Characterizing the Stylistic Fingerprint of a Frontier Language Model

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Abstract

Large language models are widely believed to have recognizable writing styles, but most popular claims (e.g. that models overuse the word *delve*) are derived from *ChatGPT* and applied indiscriminately to all models. We ask what, specifically, characterizes the prose of one frontier model—Anthropic’s Claude Opus 4.8—relative to human writing and to seven other models, holding topic constant via parallel corpora. Combining interpretable stylometry, log-odds n -gram selection, a length-controlled regression, an interpretable classifier, and suffix-automaton (n -gram novelty) mining, we find a stable fingerprint that is *structural, rhythmic, and interactional* rather than a matter of stereotyped vocabulary. Claude is identifiable from prose alone (prose-only classifier AUC 0.95 vs. a human/GPT-4o/GPT-3.5 panel; 0.87 vs. six modern models), driven by markedly higher sentence-length variability, denser content-word usage, a strong em-dash habit, and—most diagnostically—a verbatim offer-to-continue closing move (“would you like me to go deeper into...”) that, in our corpus, appears dozens of times in Claude and *zero* times across humans and all other models. We further show that the stereotyped “delve” vocabulary is a GPT (and other-model) trait that Claude *avoids*: across seven models in an essay genre, Claude is the only model that never writes *delve* and uses *crucial* least. We release the fully provenanced corpus and pipeline. Sections on the effect of reasoning effort, Claude versions, and multi-turn dynamics are forthcoming.

1 Introduction

The claim that text “sounds like AI” is now commonplace, and a folk vocabulary of tells—*delve*, *tapestry*, *crucial*, *intricate*—circulates

widely. Two problems undercut these claims as characterizations of any *particular* model. First, the evidence base is overwhelmingly about ChatGPT (Kobak et al., 2025), yet the conclusions are applied to “LLMs” generically. Second, most analyses do not control topic: a classifier that distinguishes model from human text may be learning the prompt distribution, not the writing style.

We study a single frontier model, Anthropic’s Claude Opus 4.8, and ask a deliberately narrow question: *holding the prompt fixed, what makes Claude’s prose identifiable relative to humans and to other models?* We build two topic-matched parallel corpora in which the same prompts are answered by Claude, by humans, and by up to six other models, and we attack the question with five complementary methods. Our contributions:

1. A parallel-corpus protocol and a fully provenanced dataset (every generation stored with model, decoding configuration, token usage, and timestamp) that supports reuse without regeneration.
2. Evidence that Claude’s fingerprint is *structural and rhythmic* (sentence-length burstiness, content-word density, em-dashes) and survives both markdown stripping and length control.
3. Identification of a verbatim *interactional* signature—an offer-to-continue closer—via log-odds and suffix-automaton mining, present in Claude and absent from all other sources in our corpus.
4. A reconciliation of the “delve” folklore: the stereotyped vocabulary is a GPT/other-model trait; Claude is the cleanest frontier model on it.
5. A *coverage taxonomy* (intent $\times$ inter-

083	action structure $\times$ stance) and a re-	Stylometry and n-gram novelty. Our	132
084	reproducible audit showing that standard	features build on classical stylometry, no-	133
085	corpora occupy a narrow slice (single-	tably function-word frequencies and Burrows's	134
086	turn, cooperative, informational); we ar-	Delta (Burrows, 2002). To probe memoriza-	135
087	gue that extending model characteriza-	tion versus genuine novelty, we adopt suffix-	136
088	tion into the under-observed cells—multi-	automaton n -gram search (Merrill et al., 2024)	137
089	turn corrective, contested-stance, sensi-	and unbounded n -gram indexing (Liu et al.,	138
090	tive, emotional—is itself a contribution,	2024), applying novelty mining to a single	139
091	and we release the audit tooling.	model's outputs comparatively.	
092	2 Related Work	Sycophancy and persona. Frontier mod-	140
093	Detecting machine-generated text. A	els exhibit measurable behavioral signatures:	141
094	large literature aims to decide <i>whether</i> a text is	sycophancy and agreement under pressure	142
095	machine-generated. Supervised detectors fine-	(Perez et al., 2022; Sharma et al., 2023), and	143
096	tune classifiers such as RoBERTa on model out-	self-interaction attractors such as the “spiritual	144
097	puts (Solaiman et al., 2019), while zero-shot	bliss” state reported for Claude (Anthropic,	145
098	methods exploit statistical signatures of the	2025). These motivate our future-work agenda	146
099	generating distribution: probability curvature	on multi-turn corrective behavior, where stylistic	147
100	(Mitchell et al., 2023) or the cross-perplexity	fingerprints may interact with concession	148
101	of a model pair (Hans et al., 2024). Dugan	dynamics.	149
102	et al. (2024) show such detectors are brittle	Data. We draw on parallel human/LLM cor-	150
103	across domains and attacks, and watermarking	pora including HC3 (Guo et al., 2023) and Al-	151
104	(Kirchenbauer et al., 2023) sidesteps detection	pacaEval (Li et al., 2023), repurposing them	152
105	by injecting a signal at generation time. We	for comparative stylometric characterization	153
106	do not detect machine text; we instead <i>charac-</i>	rather than detection or preference ranking.	154
107	<i>terize</i> the style of one frontier model, assuming	3 Corpus	155
108	provenance is known.	We construct two parallel tracks (Table 1). In	156
109	Attributing text to a model. Closer to	each, a fixed set of prompts is answered by	157
110	our aim is work on <i>which</i> model wrote a text.	multiple sources, so that any feature separa-	158
111	Surveys frame this as authorship attribution	ting Claude reflects <i>how</i> it writes rather than	159
112	in the LLM era (Huang et al., 2024), and dis-	<i>what</i> it was asked. Claude (and GPT-4o on the	160
113	criminative classifiers can separate model fam-	HC3 track) are generated by us with adaptive	161
114	ilies. We complement attribution: rather than	thinking at effort=high ; all other cells are	162
115	optimizing a label, we use an interpretable,	reused from the source datasets. Every record	163
116	length-controlled regression and log-odds n -	stores full provenance—model and version, de-	164
117	gram selection (Monroe et al., 2008) to surface	coding configuration, token usage, timestamp,	165
118	<i>what</i> distinguishes Claude, and we audit which	and code commit—plus the verbatim API re-	166
119	stylistic claims hold up under corpus coverage.	sponse. The seven distinct non-Claude mod-	167
120	Excess vocabulary and “AI-ese.” Kobak	els are GPT-3.5-era ChatGPT, GPT-4o, GPT-	168
121	et al. (2025) quantify LLM-marker words (e.g.	4-turbo, Gemini-pro, DeepSeek-67B, Qwen2-	169
122	“delve”) as excess vocabulary in the biomed-	72B, and Llama-3-70B; with the human an-	170
123	ical literature; popular accounts trace the habit	chor this gives eight comparison sources (six of	171
124	to RLHF annotation pipelines (Willison, 2024).	which—the “modern models”—appear on the	172
125	Such word-lists, however, are largely a Chat-	AlpacaEval track).	173
126	GPT/GPT signature. We treat the “delve”	3.1 Corpus statistics	174
127	folklore as a hypothesis to test rather than	Table 2 summarizes volume and length. Re-	175
128	a given: our markers are discovered data-	sponses are model-typical in length (Claude	176
129	drivenly from parallel corpora, and we show	median 230–267 words); the human HC3 an-	177
130	this trait is <i>not</i> characteristic of Claude.	swers are much shorter and more variable	178
		(median 77, mean 120), which we account	179

Track	Prompts	Sources (same prompt each)
HC3 (human-anchored)	200 (5 dom.)	human, ChatGPT (GPT-3.5 era), GPT-4o [†] , Claude Opus 4.8[†]
Alpaca Eval (modern)	200 (instr.)	gpt-4-turbo, gpt-4o, gemini-pro, deepseek-67b, Qwen2-72B, Llama-3-70B, Claude Opus 4.8[†]

Table 1: The two parallel corpus tracks. [†] = self-generated; all other cells reused. HC3 carries a human anchor; AlpacaEval carries the modern multi-model contrast (no human, by design). HC3 spans five source domains (reddit-ELI5 and open-QA are merged into “general” for reporting).

Track	Prompts	Prompt wds (md)	Claude resp. wds (md)	Total words
HC3	200	10	267	156k
AlpacaEval	200	18	230	354k

Table 2: Corpus volume and length (median word counts). Totals sum all sources in the track. Full per-source distributions: Appendix B.

for via length controls (§5.3). Prompts are short (median 10–18 words) and roughly half are phrased as questions. Per-domain volume and full distributions are in Appendix B. Two corpus-level register signals already foreshadow the results: Claude’s responses are markedly more *interrogative* at the sentence level (17% of sentences on HC3, 10% on AlpacaEval, vs. $\leq 4\%$ for every other source—the offer-closer, §5.4), and least readable (Flesch ≈ 40 vs. 46–52), consistent with higher content-word density.

3.2 Coverage and its limits

Because style is context-dependent, we audit *what behavior the prompts elicit*, labeling every prompt by intent, speech act, register, tone, and topic, plus interaction flags (`src/analyze_coverage.py`; Table 3). Labels are produced by Claude Opus 4.8 itself (the model under study), a circularity we flag in Limitations.

TODO (human validation pending): hand-annotate a ~ 50 -prompt sample, report inter-annotator agreement with the LLM labels, and add a small validation table. Quick human pass to be added.

The corpus occupies a narrow region of the behavior space along several axes:

- **Mood and speech act.** We measure both the *syntactic mood* (rule-

based, from punctuation and sentence-initial cues) and the *pragmatic speech act* (LLM-labeled), which differ because many syntactically interrogative prompts (“Can you write...?”) are pragmatic directives. Syntactically, prompts are 74%/58% (HC3/AlpacaEval) interrogative, 2%/20% imperative, $\sim 22\%$ declarative; pragmatically they are questions or directives with ≤ 1 *assertion* per track (Appendix B). Either way, the corpus almost never presents Claude with an assertion to agree with, qualify, or contest.

- **Interaction.** Only $\sim 2\%$ of prompts imply multi-turn interaction, and essentially none are adversarial or corrective.
- **Intent.** Code, creative, opinion, and emotional intents are rare or absent, especially in HC3 (Table 3).
- **Stance / register / tone.** Prompts are predominantly neutral in tone and information-seeking; subjective or contested-stance prompts are a small minority.

Consequently the fingerprint we report should be read as Claude’s *single-turn informational* voice. Behaviors gated on other cells—e.g. the concession move “you’re right to push back on that,” which requires a *declarative* challenge across turns—are out of scope here and motivate the corpus expansion (more prompt variety across topic, register, tone, speech act, and interaction structure) that we describe as future work (§5.8, Appendix B).

4 Methods

We compute interpretable features per response (lexical: type-token ratio, hapax rate, function-word rate; punctuation: em-dash, colon, question, exclamation, emoji; syntax: mean sentence length and *burstiness*, the ratio of the standard deviation to the mean of words-per-sentence; markdown/structure; and rhetoric). Rates are per 100 or 1000 words so that text length does not mechanically dominate. We compute two dozen such features per response. We summarize source differences with Cohen’s *d* (Claude vs. a comparison group); all reported effects are significant at $p < 0.001$ (Welch’s *t*, $n=200$

Intent (primary)	HC3	AlpacaEval
explanation	94	33
factual QA	53	29
advice/recommendation	49	31
instruction task	3	26
creative writing	0	24
math/reasoning	0	14
code	0	11
opinion/subjective	0	11
other (classif., edit, roleplay)	1	20
<i>flag</i> : multi-turn implied	8	4
<i>flag</i> : subjective	14	39
<i>flag</i> : sensitive	61	16

Table 3: Prompt intent distribution and interaction flags (HC3 $n=200$; AlpacaEval $n=199$, one prompt unlabeled). Single frozen audit run (`coverage_labels.csv`). The corpus is single-turn, cooperative, and information-oriented.

Claude vs. ≥ 200 comparison). The classifier is ℓ_2 -regularized logistic regression ($C=1.0$, balanced class weights). To separate *voice* from *formatting*, we additionally compute all prose features after stripping markdown markup. We mine characteristic n -grams with the Monroe et al. (2008) log-odds estimator; quantify length-independence with OLS (§5.3); measure separability with ℓ_2 -regularized logistic regression under 5-fold cross-validation; and mine arbitrary-length spans and n -gram novelty with a suffix automaton (Merrill et al., 2024).

5 Results

5.1 A large signal, dominated at first by formatting

The strongest raw separators between Claude and the other HC3 sources are markdown-structural: markdown headers ($d=2.9$), bold ($d=2.3$), and bullets ($d=1.8$), which the human and old-ChatGPT cells use essentially never. This is real but is a *formatting* habit; the decisive question is whether anything survives removing it.

5.2 The voice survives markdown stripping

With all markup stripped, Claude still separates strongly on prose (Table 4): it interleaves short and long sentences far more than any other source (burstiness), writes denser content-word prose (lower function-word rate), and retains a marked em-dash habit. These

Prose feature (markdown stripped)	$ d $	Claude vs. others
Sentence burstiness	1.69	0.86 vs. 0.36–0.55
Function-word rate (lower)	1.47	0.33 vs. 0.37–0.44
Em-dash rate	0.87	0.88 vs. ≤ 0.40
Colon rate	0.84	1.41 vs. 0.24–1.69
Question rate	0.82	0.61 vs. 0.01–0.28

Table 4: HC3 track, Claude vs. pooled others, prose only. Mean |Cohen’s d | and group means.

are the load-bearing stylistic findings.

5.3 The signature is not a length artifact

On the HC3 track, Claude writes longer answers (mean 253 words vs. 120 for humans, 228 for GPT-4o), so we test whether the prose effects merely track length. Within every word-count quartile the effects persist, and a length-controlled regression (HC3) (feature $\sim$ is_claude + $z(n_words)$) leaves the Claude coefficient large and highly significant: burstiness $t=20.6$, function-word rate $t=-14.4$, em-dash $t=12.6$, question rate $t=4.6$. Type-token ratio is length-sensitive and we treat lexical-diversity claims as length-conditioned.

5.4 The interactional signature: an offer-to-continue closer

Data-driven n -gram mining (Monroe et al., 2008) surfaces a clear interactional fingerprint rather than a content-word list. The most Claude-like trigrams (HC3) are *would you like*, *like me to*, *me to explain*, *in more detail*, *to go deeper*; the anti-signature—terms Claude *underuses*—is led by the formal frame *it is important to*. At the sentence level this closer registers as a striking jump in interrogatives: **17%/10% of Claude’s response sentences are questions (HC3/AlpacaEval), versus $\leq 4\%$ for every other source** (Appendix B). Suffix-automaton mining (§5.7) confirms the pattern at arbitrary length.

5.5 The “delve” vocabulary is a GPT trait Claude avoids

On the AlpacaEval track—an essay/instruction genre that *does* elicit the stereotyped words—Claude is the cleanest of seven models (Table 5). It is the only model that never writes *delve* or *delves*, and it uses *crucial* least of all seven (0.039 per 1k vs. DeepSeek 0.054, Gemini 0.090,

per-1k	Claude	g4t	g4o	qwen	llama
delve	0	.018	.018	0	.017
delves	0	.037	.089	.019	.034
crucial	.039	.422	.340	.346	.185
intricate	0	0	0	.019	.050
pivotal	0	.073	.018	.038	0

Table 5: Frequency (per 1000 words) of stereotyped “excess” words on the AlpacaEval essay track. g4t = gpt-4-turbo, g4o = gpt-4o. Claude is at or near the minimum for every term. (gemini/deepseek omitted for space; same pattern.)

Track	with formatting	prose only
HC3 (vs. human/GPT-4o/old-ChatGPT)	0.982	0.946
AlpacaEval (vs. 6 modern models)	0.901	0.867

Table 6: Claude-vs-rest ROC-AUC, logistic regression, 5-fold CV.

Llama 0.185, GPT-4o 0.340, Qwen 0.346, GPT-4-turbo 0.422). Claude’s own lexical lean is conversational (“actually” occurs $\sim 6\times$ more often in Claude than GPT-4o on HC3); this conversational lean is a secondary signal beneath the structural one.

5.6 Separability

An interpretable classifier confirms a stable, learnable voice (Table 6). Stripping mark-down reduces AUC only modestly, so the classifier is not merely reading layout; and its top coefficients are exactly the features of §5.2 (burstiness, function-word density, em-dash, question rate), an independent-method confirmation. Raw length is not a feature, and residualizing every feature on response length barely changes separation (prose-only AUC $0.946 \rightarrow 0.931$ on HC3, $0.867 \rightarrow 0.861$ on AlpacaEval), so the result is not a length artifact.

5.7 Arbitrary-length spans and n -gram novelty

Suffix-automaton mining (Merrill et al., 2024) gives the most concrete form of the signature. Within each track, spans up to *eight words* occur dozens of times in Claude and *zero* times across that track’s non-Claude sources (HC3: humans + 2 models; AlpacaEval: 6 models, no human): on HC3, *would you like me to explain any* ($\times 20$) and *would you like me to go deeper into* ($\times 14$); on AlpacaEval, *would you*

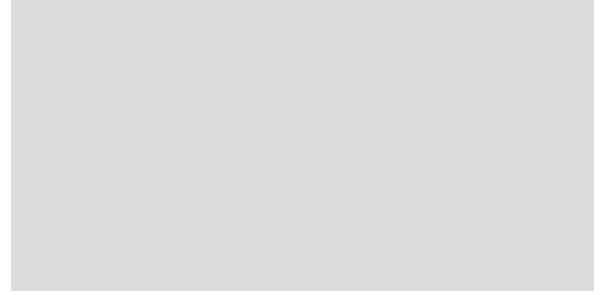


Figure 1: **[Placeholder image—replace with real plot.]** n -gram novelty by source: the fraction of each source’s n -grams ($n=1\dots 10$) that are absent from the pooled other sources (AlpacaEval track). Claude’s curve sits above all six other models at every n (bigram novelty 0.519 vs. 0.27–0.36), i.e. its phrasing overlaps least with the pool. Data: reports/ngram_dawg.md.

Replace the grey placeholder with the real novelty-curve figure (one line per source).

like me to expand on any ($\times 7$). Claude also has the highest n -gram novelty of any source (Figure 1; e.g. bigram novelty 0.519 vs. 0.27–0.36 for the six AlpacaEval models): its exact phrasing overlaps least with the pool. We read novelty as relative distinctiveness, not absolute creativity (see Limitations).

5.8 Secondary analyses (in progress)

Three planned analyses are not yet included; designs and placeholder tables appear in Appendix C.

- **Effect of reasoning effort/thinking.** Whether the fingerprint intensifies or attenuates across `effort` $\in$ {low, medium, high} and with thinking disabled. *[To be completed; Appendix C.1.]*
- **Across Claude versions.** Intra-Claude drift across model versions (e.g. Opus/Sonnet/Haiku and older releases). *[To be completed; Appendix C.2.]*
- **Multi-turn and self-interaction.** Turn-dependent traits and the Claude-to-Claude “spiritual bliss” attractor reported in (Anthropic, 2025). *[To be completed; Appendix C.3.]*

6 Discussion

Across two corpora, five methods, and eight comparison sources, the same fingerprint recurs and survives every control: Claudes-

peak is markdown-scaffolded, rhythmically varied, content-dense, and engagement-and-offer-oriented prose—and specifically *not* the “delve” register. A methodological lesson follows: borrowed AI-word lists largely measure *GPT*, and characterizing a specific model requires data-driven discovery against topic-matched references.

Limitations

We study a single Claude configuration (Opus 4.8, `effort=high`); the effort, version, and multi-turn axes are not yet measured (§5.8). Crucially, our coverage audit (§3.2) shows the corpus is single-turn, cooperative, and information-oriented, so the reported fingerprint is Claude’s *single-turn informational* voice; context-gated behaviors (corrective concession, refusal, emotional support, code/creative registers) are out of scope and are the target of the planned expansion (English only). Reused contrast completions are their published vintages (e.g. gemini-pro 1.0, Qwen2-72B, GPT-4-turbo/4o 2024), modernish but not the absolute latest. *n*-gram novelty is measured relative to our corpus pool, not a large external reference, and may be inflated for Claude as the lone out-of-cluster (and only freshly generated) source. The HC3 human anchor is informal web text, so part of the human–Claude gap reflects register rather than authorship. The coverage audit’s labels are produced by an LLM (the model under study), introducing a circularity; a human-validation pass on a labeled sample is pending. Finally, raw length is excluded from the classifier features and the feature effects are length-controlled (§5.3), but some prose features still correlate with length; we report a length-residualized classifier as a check (§5.6).

Data and Reproducibility

We do not test mechanism, but the interactional, low-formality, offer-oriented profile is consistent with post-training (RLHF) that rewards helpful, engaging, and appropriately hedged responses. Code and the fully provenanced corpus (generations plus metadata, and the audit tooling) will be released at an anonymized URL. HC3 and AlpacaEval are public research artifacts used under their li-

censes; we redistribute only derived statistics and our own model generations.

Ethics Statement

This work characterizes model writing style using public datasets and model outputs; it involves no human subjects or private data. Stylistic fingerprints can aid provenance and detection but could also inform evasion; we report aggregate, interpretable features rather than a deployable detector.

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527	Bowman, Newton Cheng, Esin Durmus, Zac	the lowest Flesch (≈ 40 , denser/harder) of all	578
528	Hatfield-Dodds, Scott R. Johnston, Shauna	sources on both tracks.	579
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530	Kamal Ndousse, Oliver Rausch, Nicholas	Prompt variety (audit, counts of 200	580
531	Schiefer, Da Yan, Miranda Zhang, and Ethan	per track). The prompts are narrow on ev-	581
532	Perez. 2023. Towards understanding sycop-	ery axis (Table 7): almost all are <i>questions</i>	582
533	hancy in language models. <i>arXiv preprint</i>	or <i>directives</i> (assertions ≤ 1); tone is over-	583
534	<i>arXiv:2310.13548</i> .	whelmingly <i>neutral</i> (82–87%); register is neu-	584
535	Irene Solaiman, Miles Brundage, Jack Clark,	tral/casual with almost no <i>formal</i> prompts;	585
536	Amanda Askell, Ariel Herbert-Voss, Jeff Wu,	and topics concentrate (HC3: health 28%, sci-	586
537	Alec Radford, Gretchen Krueger, Jong Wook	ence 24%, finance 20%). This uniformity—not	
538	Kim, Sarah Kreps, Miles McCain, Alex New-	insufficient sample size—is the coverage prob-	
539	house, Jason Blazakis, Kris McGuffie, and Jas-	lem the expansion targets.	
	mine Wang. 2019. Release strategies and the so-		
	cial impacts of language models. <i>arXiv preprint</i>		
	<i>arXiv:1908.09203</i> .		

Axis	Value	HC3	Alpaca
speech act	question	150	106
	directive	46	93
	assertion	1	0
register	neutral	130	130
	casual	70	62
	formal	0	7
tone	neutral	165	175
	emotional	20	5
	playful / urgent	1/14	18/1

Table 7: Prompt variety along *pragmatic* speech act (LLM-labeled), register, and tone (HC3 $n=200$; AlpacaEval $n=199$). Assertions, formal register, and non-neutral tones are nearly absent. Topic and intent breakdowns: `coverage_labels.csv`. (Syntactic mood, which differs, is in §3.2.)

C.3 Multi-turn and self-interaction

Planned design. (i) Two-turn parallel probes (e.g. MT-Bench-style) to measure turn-dependent traits comparatively; (ii) elicited Claude-to-Claude conversations to test the “spiritual bliss” attractor (Anthropic, 2025) quantitatively (e.g. trajectory of consciousness/gratitude vocabulary and vocabulary collapse over turns). *[Results table to be inserted.]*

Coverage expansion (planned). To move beyond the single-turn informational slice (§3.2) we will (i) sample real first-turn prompts from WildChat / LMSYS-Chat-1M to populate under-covered intents (code, creative, opinion, emotional); (ii) *construct* multi-turn corrective scenarios (user challenges or corrects Claude) to elicit concession / push-back behavior; and (iii) add a small, defensively-framed sensitive/refusal set. Targets are ~ 100 – 150 prompts per intent cell and ~ 80 – 120 corrective scenarios; the audit tooling re-runs on the expanded corpus.

C Placeholders for secondary analyses

C.1 Effect of reasoning effort

Planned design. Regenerate the 200 HC3 prompts (and 200 AlpacaEval prompts) under $\text{effort} \in \{\text{low}, \text{medium}, \text{high}\}$ and with thinking disabled, holding all else fixed; recompute the §5.2 features and the §5.6 classifier within each setting; report whether burstiness, density, em-dash, and the offer-closer scale with effort. *[Results table to be inserted.]*

C.2 Across Claude versions

Planned design. Generate the same prompts with additional Claude versions (e.g. Sonnet/Haiku, and an older 3.x), then measure intra-Claude drift in the fingerprint features and the offer-closer rate. *[Results table to be inserted.]*